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United States Patent	12404077
Kind Code	B2
Date of Patent	September 02, 2025
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Flexible enclosure having a seal and method and system for making the same

Abstract

Flexible enclosure includes a front wall and a back wall that oppose each other and a side wall that joins the front and back walls. The front wall, the back wall, and the side wall are discrete sections. The front wall and the back wall are attached to opposite edges of the side wall. The side wall is configured to have a slit between top and bottom wall portions of the side wall. The flexible enclosure also includes a resealer extending between the front and back walls along the bottom wall portion of the side wall. The bottom wall portion is separable from the top wall portion along the slit. The bottom wall portion is configured to flex outwardly and away from the top wall portion and form a spout of the flexible enclosure. The resealer is configured to close the spout.

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Appl. No.:	17/186076
Filed:	February 26, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210269196 A1	Sep. 02, 2021

Related U.S. Application Data

us-provisional-application US 62983892 20200302

Publication Classification

Int. Cl.: B65D33/25 (20060101); B65D30/20 (20060101); B65D75/58 (20060101)

U.S. Cl.:

CPC B65D33/2508 (20130101); B65D31/10 (20130101); B65D75/5827 (20130101); B65D75/5861 (20130101);

Field of Classification Search

CPC: B65D (33/2508); B65D (31/10); B65D (75/5827); B65D (75/5861)

USPC: 383/42

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6971794	12/2004	Yeager	N/A	N/A
7442156	12/2007	Yeager	N/A	N/A
8182407	12/2011	Yeager	N/A	N/A
9962898	12/2017	Russell et al.	N/A	N/A
2004/0208402	12/2003	Yeager	383/120	B65D 75/5822
2007/0086683	12/2006	Yeager	N/A	N/A
2007/0258665	12/2006	Greco	383/63	B65D 33/2508
2008/0008404	12/2007	Brauer	383/61.1	B65D 33/2508
2008/0008406	12/2007	Russell	383/63	B65D 75/5866
2013/0071047	12/2012	VanLoocke	53/469	B65D 33/16
2017/0057192	12/2016	Romenesko	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20115181	12/2001	DE	B65D 31/10
187308	12/2007	EP	N/A
1873081	12/2007	EP	B31B 19/36
2535287	12/2011	EP	B65D 33/24
3715101	12/2019	EP	B31B 70/8132
2816597	12/2001	FR	B65D 75/5816

OTHER PUBLICATIONS

Machine translation of FR-2816597-A1. cited by examiner

Machine translation of DE-20115181-U1. cited by examiner

Machine translation of EP-1873081-A1. cited by examiner
Machine translation of EP-2535287-A1. cited by examiner
Search Report for corresponding European Patent Application No. 21159872.7 mailed Jul. 6, 2021 (7 pages). cited by applicant
EPO Exam Report dated Oct. 16, 2024 for corresponding European Patent Application No. 21159872.7 (7 pages). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims the benefit of U.S. Provisional Application No. 62/983,892, filed on 2 Mar. 2020, which is incorporated herein by reference in its entirety.

FIELD

(1) The subject matter of the present application relates to flexible, resealable enclosures and methods of manufacturing the same.

BACKGROUND

(2) Flexible enclosures (e.g., box lines, plastic packages, and the like) can be used to hold a wide variety of contents. Flexible enclosures have become more popular because such enclosures can cost less than alternatives and can be readily opened but also sealed to provide sufficient protection of the contents therein. A resealable enclosure enables a user to repeatedly open the enclosure, remove a portion of the contents from the enclosure, and then close the enclosure in a manner that seals the contents therein. In many cases, these resealable enclosures are torn or ripped when initially opened. Despite being irreparably altered, the resealable enclosures are configured to reseal the opening to protect the contents from the surrounding environment and/or prevent the contents from inadvertently exiting the enclosure. Compared to user-improvised methods (e.g., rolling the top of a flexible bag), resealable enclosures are tidier and reassure the user that the contents are protected and will not spill if the enclosure is mispositioned or dropped.

(3) One example of a flexible enclosure can form a spout through which the contents may be poured. For example, a corner of a flexible enclosure may be removed that allows a gusseted edge to be pulled outwardly and form a spout. In this design, however, the flexible enclosure is formed from a single section of film. More specifically, a single roll of film may be unwound so that opposite longitudinal edges of the film extend parallel to a central axis. A section of the film may be cut transversely with respect to the central axis, thereby forming a bottom transverse edge that is opposite a top transverse edge of the section. The section may be wrapped around the central axis so that the longitudinal edges overlap themselves and the section forms opposite rounded sides. These rounded sides are pushed inward to form gusseted sides. The top transverse edge may be sealed to itself and the bottom transverse edge may be sealed to itself.

(4) Although one of the gusseted sides is capable of forming the spout, the single section of film limits modifications to the design. Alternative designs are desired.

BRIEF DESCRIPTION

(5) In an embodiment, a flexible enclosure is provided that includes a front wall and a back wall that oppose each other and a side wall that joins the front and back walls. The front wall, the back wall, and the side wall are discrete sections. The front wall and the back wall are attached to opposite edges of the side wall. The side wall is configured to have a slit between top and bottom

wall portions of the side wall. The flexible enclosure also includes a resealer extending between the front and back walls along the bottom wall portion of the side wall. The bottom wall portion is separable from the top wall portion along the slit. The bottom wall portion is configured to flex outwardly and away from the top wall portion and form a spout of the flexible enclosure. The resealer is configured to close the spout.

(6) Optionally, the side wall is configured to have a line of weakness, the slit being formed when the line of weakness is broken.

(7) Optionally, the side wall is a gusseted side wall.

(8) Optionally, the resealer includes a self-mating strip extending between the front and back walls along an interior surface of the side wall. The self-mating strip may have opposite crushed ends and each edge of the side wall may have one of the crushed ends.

(9) Optionally, the resealer includes first and second segments that are configured to mate with one another to close the spout.

(10) Optionally, the side wall is a first side wall and the flexible enclosure includes a second side wall that is also a discrete section with respect to the front wall, the back wall, and the first side wall. The front wall and the back wall are attached to opposite edges of the second side wall, and the first and second side walls oppose each other.

(11) Optionally, the resealer is a first resealer and the second side wall has a second resealer that extends between the front and back walls. The second side wall is configured to have a slit between top and bottom wall portions of the second side wall, wherein the second resealer extends between the front and back walls along the bottom wall portion of the second side wall. The bottom wall portion of the second side wall is separable from the top wall portion along the slit. The bottom wall portion of the second side wall is configured to flex outwardly and away from the top wall portion. The resealer of the second side wall is configured to close the second side wall.

(12) Optionally, the front and back walls have respective slits and respective sealers. The slits of the first and second side walls and the front and back walls enable a user to remove a top of the flexible enclosure to provide an open-ended enclosure having an open top. The sealers of the first and second side walls and the front and back walls enable a user to close the open top of the open-ended enclosure.

(13) Optionally, the resealer is one of four sealers. Each of the sealers extends along a respective one of the first side wall, the second side wall, the front wall, and the back wall.

(14) In an embodiment, a method is provided that includes providing a first continuous web of flexible material moving in a first flow direction. The first continuous web has spaced-apart fastener strips that extend transverse to the first flow direction. The method also includes separating a first segment from the first continuous web and folding the first segment such that at least one of the fastener strips is folded over itself, thereby forming a folded segment. The method also includes providing second and third continuous webs of flexible material that includes second and third segments, respectively. The method also includes coupling the second and third segments to each other and to opposite edges of the folded segment, thereby at least partially forming a flexible enclosure that includes a front wall, a back wall, and a side wall formed by the folded segment. The front and back walls are coupled by the side wall with the fastener strip extending between the front and back walls.

(15) Optionally, the side wall is configured to have a slit between top and bottom wall portions of the side wall. The slit permits the bottom wall portion to flex outwardly and away from the top wall portion and form a spout of the flexible enclosure. The fastener strip is configured to close the spout.

(16) Optionally, providing the first continuous web of flexible material includes the first continuous web having spaced apart lines of weakness that extend transverse to the first flow direction. The lines of weaknesses extend along and operably near respective ones of the fastener strips, wherein the top and bottom wall portions are separable when the line of weakness is broken.

(17) Optionally, the side wall is a gusseted side wall.

(18) Optionally, the fastener strip of the side wall extends along an interior surface of the side wall.

(19) Optionally, the fastener strips extend along at most half of a width of the first continuous web.

(20) Optionally, the fastener strips extend essentially across an entire width of the first continuous web.

(21) Optionally, at least one of the second or third continuous webs has a continuous fastener strip that extends parallel to a flow direction of the at least one second or third continuous webs.

(22) In an embodiment, a system is provided that includes a first actuator configured to unwind a first continuous web of flexible material. The first continuous web moves in a first flow direction. The system also includes a zipper applicator configured to apply fastener strips to the first continuous web. The fastener strips extend transverse to the first flow direction. The system also includes a contoured surface configured to fold the flexible material of the first continuous web such that at least one of the fastener strips is folded over itself. The system also includes a cutter configured to separate a folded first segment from the first continuous web and at least one actuator configured to unwind second and third continuous webs of flexible material that include second and third segments, respectively. The system also includes a heat-sealing assembly configured to couple the second and third segments to each other and to opposite edges of the folded first segment, thereby at least partially forming a flexible enclosure that includes a front wall, a back wall, and a side wall formed by the folded segment. The front and back walls are coupled by the side wall with the fastener strip extending between the front and back walls.

(23) Optionally, the cutter is configured to separate another segment from the first continuous web, the other segment forming a second side wall of the flexible enclosure.

(24) In an embodiment, a flexible enclosure is provided that includes a first wall and a second wall that oppose each other and a side wall that joins the first and second walls. The first wall, the second wall, and the side wall are discrete sections. The first wall and the second wall are attached to opposite edges of the side wall. At least one of the first wall, the second wall, or the side wall is configured to have a slit that separates top and bottom wall portions. The flexible enclosure also includes a seal extending along the slit of the at least one first wall, second wall, or side wall and coupling the top and bottom wall portions. The bottom wall portion is separable from the top wall portion along the seal. The bottom wall portion is configured to flex outwardly and away from the top wall portion.

(25) Optionally, the at least one first wall, second wall, or side wall is configured to have a line of weakness. The slit is formed when the line of weakness is broken.

(26) Optionally, the seal is a one-time frangible seal that can be opened cohesively or adhesively or is a resealer configured to be sealably closed after opening.

(27) Optionally, the resealer includes a self-mating strip extending between the front and back walls along an interior surface of the side wall.

(28) Optionally, the first wall includes the slit and the seal.

(29) In an embodiment, a method is provided that includes providing a first continuous web of flexible material moving in a first flow direction and separating a first segment from the first continuous web and folding the first segment such that at least a portion of the first segment is folded over itself, thereby forming a folded segment. The method also includes providing second and third continuous webs of flexible material that includes second and third segments, respectively. The second and continuous webs move in first and second flow directions. The method also includes coupling the second and third segments to each other and to opposite edges of the folded segment, thereby at least partially forming a flexible enclosure that includes a front wall, a back wall, and a side wall formed by the folded segment. The first flow direction is transverse to at least one of the first flow direction or the second flow direction. At least one of the first continuous web has spaced-apart fastener strips that extend transverse to the first flow direction, the second continuous web has a continuous fastener strip that extends parallel to the second flow

direction, the third continuous web has a continuous fastener strip that extends parallel to the third flow direction.

(30) Optionally, at least two of the following is true: The first continuous web has spaced-apart fastener strips that extend transverse to the first flow direction. The second continuous web has a continuous fastener strip that extends parallel to the second flow direction. The third continuous web has a continuous fastener strip that extends parallel to the third flow direction.

(31) Optionally, the first continuous web has spaced-apart fastener strips that extend transverse to the first flow direction, the second continuous web has a continuous fastener strip that extends parallel to the second flow direction, and the third continuous web has a continuous fastener strip that extends parallel to the third flow direction.

(32) Optionally, the spaced-apart fastener strips and/or the continuous fastener strip are one-time frangible seals that can be opened cohesively or adhesively or are resealers configured to be sealably closed after opening.

(33) In an embodiment, a system is provided that includes a first actuator configured to unwind a first continuous web of flexible material. The first continuous web moves in a first flow direction. The system also includes a second actuator configured to unwind a second continuous web of flexible material. The second continuous web moves in a second flow direction. The system also includes a third actuator configured to unwind a third continuous web of flexible material. The third continuous web moves in a third flow direction. The system also includes at least one zipper applicator configured to apply a fastener strip to at least one of the first continuous web, the second continuous web, or the third continuous web. The system also includes a contoured surface configured to fold the flexible material of the first continuous web such that at least a portion of a first segment is folded over itself, thereby forming a folded segment. The system also includes a cutter configured to separate the folded segment from the first continuous web. The first flow direction is transverse to at least one of the first flow direction or the second flow direction. The at least one zipper applicator at least one of applies spaced-apart fastener strips to the first continuous web that extend transverse to the first flow direction, applies a continuous fastener strip to the second continuous web that extends parallel to the second flow direction, or applies a continuous fastener strip to the third continuous web that extends parallel to the third flow direction.

(34) Optionally, at least two of the following are true. The at least one zipper applicator applies spaced-apart fastener strips to the first continuous web that extend transverse to the first flow direction. The at least one zipper applicator applies a continuous fastener strip to the second continuous web that extends parallel to the second flow direction, or the at least one application applies a continuous fastener strip to the third continuous web that extends parallel to the third flow direction.

(35) Optionally, the spaced-apart fastener strips and/or the continuous fastener strip are one-time frangible seals that can be opened cohesively or adhesively or are resealers configured to be sealably closed after opening.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The inventive subject matter will now be illustrated with reference to the following figures, in which:

(2) FIG. 1 is an isolated perspective view of a flexible enclosure formed in accordance with an embodiment;

(3) FIG. 2 is a cross-section of the flexible enclosure of FIG. 1 prior to the flexible enclosure being opened;

(4) FIG. 3 is a cross-section of the flexible enclosure of FIG. 1 after being opened but prior to being

resealed;

(5) FIG. 4 is a cross-section of the flexible enclosure of FIG. 1 after being resealed;

(6) FIG. 5 is a schematic illustration of a system formed in accordance with an embodiment that may be used to make the flexible enclosure of FIG. 1;

(7) FIG. 6 is a flowchart of a method in accordance with an embodiment;

(8) FIG. 7 is a cross-section of a three-sided flexible enclosure in accordance with an embodiment prior to being opened;

(9) FIG. 8 is a schematic illustration of a system formed in accordance with an embodiment that may be used to make the flexible enclosure of FIG. 7;

(10) FIG. 9A is a cross-section of a closed three-sided flexible enclosure in accordance with an embodiment having a two-piece resealer;

(11) FIG. 9B is a cross-section of an opened three-sided flexible enclosure in accordance with an embodiment having a two-piece resealer;

(12) FIG. 10 is a schematic illustration of a system formed in accordance with an embodiment that may be used to make the flexible enclosure of FIGS. 9A and 9B;

(13) FIG. 11 is a cross-section of a four-sided flexible enclosure in accordance with an embodiment; and

(14) FIG. 12 is a schematic illustration of a system formed in accordance with an embodiment that may be used to make the flexible enclosure of FIG. 11.

(15) FIG. 13 is an isolated perspective view of a flexible enclosure formed in accordance with an embodiment.

DETAILED DESCRIPTION

(16) One or more embodiments of the inventive subject matter set forth herein include a resealable flexible enclosure and methods of making the same. The flexible enclosure can include at least two discrete sections or segments that form walls of the flexible enclosure. An access opening can be formed, through which contents (e.g., product) may pass into or out of the flexible enclosure. For example, the access opening can be provided by a spout that is formed along one side of the flexible enclosure. Optionally, the access opening can be formed when a top portion of the flexible enclosure is removed. The access opening may be sealed and re-sealed. In particular embodiments, the flexible enclosure may be manufactured using a machine such as a Totani pouch-making machine (e.g., BH Series or FD Series machine).

(17) The sections (or segments) that form the flexible enclosure can be discrete because, prior to being attached to each other, the sections are separate from each other. The sections may be provided by at least two different continuous webs of flexible material. For example, a first section (or front wall) and a second section (or back wall) may be essentially rectangular sections having edges. An edge of the first section may be either attached to an edge of the second section (e.g., along a top or bottom of the flexible enclosure) or attached to an edge of the side wall. Likewise, an edge of the second section may be either attached to an edge of the first section (e.g., along a top or bottom of the flexible enclosure) or attached to an edge of the side wall.

(18) In the illustrated embodiment described below, the resealer includes a single self-mating fastener strip. A self-mating fastener strip includes a self-mating zipper strip, an intermeshable closure member, an adhesive reseal or the like. An intermeshable closure member may include a strip of a structured surface having ridges and troughs that can mate with itself. An intermeshable closure member may include two opposing hook-filled strips having the same or similar structure. When brought together, the hooks of one strip may couple to the hooks of the opposing strip, thereby sealing the enclosure. The hooks could be configured to engage on multiple levels. Widths of the intermeshable closure members may be configured so that precise alignment is not required.

(19) In other embodiments, however, the resealer may include two or more fastener strips or other types of resealers. For example, the resealer may include separate but parallel zipper strips, different adhesive strips, hook-and-loop fastener elements, or a slider or a combination thereof. The

resealer may include, for example, a track and a strip on opposing surfaces of the enclosure in which the strip fits within and along the track. Optionally, the resealer may include two tracks and two strips in which each track receives one strip.

(20) Optionally, the resealer may include one or more adhesive reseals. The adhesive reseal may comprise a polymeric material, such as at least one of polyethylene (including a polyethylene copolymer), polypropylene, ionomers, amorphous polyester, vinyl acetate, polybutylene, or nylon ethylene-vinyl alcohol copolymer (EVOH). In some embodiments, the adhesive seal may include a material that is capable of being reused to open and close the barrier. In other embodiments, the resealer may be substituted with a one-time frangible seal that can be opened cohesively or adhesively.

(21) In the following description and claims, relative or spatial terms such as “front,” “back,” “side,” “top,” “bottom,” “lateral,” “longitudinal,” and the like are only used to distinguish the referenced elements or features with respect to one another and make the language more readily understandable. The terms do not necessarily require particular positions, sizes, or orientations relative to the surrounding environment. Moreover, in the following description and claims, the terms “first,” “second,” and “third,” etc. may be used as labels to distinguish similar elements (e.g., first and second side walls) and are not intended to impose numerical requirements on their objects.

(22) FIG. 1 illustrates a perspective view of a resealable flexible enclosure **100**. In the illustrated embodiment, the flexible enclosure **100** includes a front wall (or first wall) **102**, a back wall (or second wall) **104**, and first and second side walls **106**, **108**. Although FIG. 1 shows two side walls **106**, **108**, the flexible enclosure **100** may include only one side wall in other embodiments.

(23) The walls **102**, **104**, **106**, **108** are discrete sections of a flexible material. In some embodiments, the discrete sections may be provided by different webs (or films) of the flexible material. Discrete sections are not portions of the same piece of flexible material. Discrete sections are typically cut from different webs or cut from the same web at different times. For example, the front wall **102** may be obtained from a first continuous web of flexible material, the back wall **104** may be obtained from a different second continuous web of the same flexible material, and the side walls **106**, **108** may be obtained from a third continuous web of the same flexible material. In some embodiments, different walls may comprise different types of material. In an alternative embodiment, the different walls may be provided by the same continuous web.

(24) In the illustrated embodiment, the front wall **102** and the back wall **104** are directly attached to each other along a top sealed region **110** and a bottom sealed region **112**. The front wall **102** is also directly attached to the first side wall **106** along a top sealed region **114** and a bottom sealed region **116**. The front wall **102** is also directly attached to the second side wall **108** along a top sealed region **118** and a bottom sealed region **120**. The back wall **104** is also directly attached to the first side wall **106** along a top sealed region **124** and the second side wall **108** along a top sealed region **126**. Although not shown, the back wall **104** is also directly attached to the second side wall **108** along a bottom sealed region and the first side wall **106** along a bottom sealed region.

(25) Also shown in FIG. 1, the first side wall **106** includes a line of weakness **132** and a resealer **134**. In the illustrated embodiment, the first and second side walls **106**, **108** are gusseted side walls such that a portion of the side walls are folded inwardly between the front and back walls **102**, **104**. In other embodiments, however, the first and second side walls **106**, **108** may not be gusseted.

(26) FIGS. 2-4 illustrate cross-sections of a portion of the flexible enclosure **100**. In the illustrated embodiment, the line of weakness **132** (FIG. 1) causes a slit **145** (FIG. 2) to be formed when the line of weakness **132** is ruptured (e.g., by a user's thumb). The slit **145** is positioned between a top wall portion **140** (FIG. 1) and a bottom wall portion **142** of the side wall **106**. In FIGS. 2-4, a profile of the top wall portion **140** is shown as a dashed line. The slit **145** becomes an access opening **150** (shown in FIG. 3) when the bottom wall portion **142** is moved outwardly. In the illustrated embodiment, the resealer **134** is a self-mating fastener strip located along an interior surface **107** of the first side wall **106**. A self-mating fastener strip includes either (a) only a single

segment that is folded onto itself when the access opening **150** is sealed or (b) two segments, having the same profile or structure, that oppose and engage each other when the access opening **150** is sealed. The self-mating fastener strip may be, for example, a self-mating zipper strip or an intermeshable closure member.

(27) In other embodiments, however, the resealer **134** may include other types of fasteners that may or may not be self-mating. For example, in one alternative embodiment, a zipper assembly may permit the side wall to be separated into top and bottom portions, thereby providing an access opening **150**, while two zipper strips having different profiles may be positioned along an interior surface of the bottom portion and engage each other.

(28) FIG. **3** illustrates how the first side wall **106** forms a spout **152**. The spout **152** corresponds to an increased dimension (or width) of the flexible enclosure **100**. When the flexible enclosure **100** includes the spout **152**, the width of the flexible enclosure **100** is greater than the width of the flexible enclosure **100** prior to forming the spout **152**. To seal or close the access opening **150**, sides of the spout **152** that are approximately parallel to the front and back walls **102**, **104** are pressed toward each other. As shown in FIG. **4**, the side wall **106** remains in an extended or projected form but the resealer **134** closes the access opening **150**. In FIG. **4**, the profile of the top wall portion **140** is exaggerated inwardly. In other embodiments, the top wall portion **140** may essentially coincide with ends of the resealer **134** such that the access opening **150** is effectively closed.

(29) Accordingly, in some embodiments, a user can open the flexible enclosure **100** by pressing into (e.g., with the user's thumb or index finger) the first side wall **106** to break the line of weakness **132**. The line of weakness **132** is between the top and bottom wall portions **140**, **142** but may be located closer to the top sealed regions **114**, **118**, **124**, **126** than the bottom sealed regions **116**, **120**. Once broken, the slit **145** may exist between the top and bottom wall portions **140**, **142**. The user may then pull the first side wall **106** away from an interior of the flexible enclosure **100**. More specifically, the user may grip the interior surface of the bottom wall portion **142** and pull the bottom wall portion **142** outwardly. An access opening **150** develops from the slit **145** as a distance between an edge of the bottom wall portion **142** and an edge of the top wall portion **140** becomes greater. The top wall portion **140** is not moved outwardly. In alternative embodiments, the slit **145** may exist between two opposing zipper strips of a zipper assembly.

(30) With the bottom wall portion **142** projecting outwardly and the top wall portion **140** remaining in position, the spout **152** is formed. Content (e.g., cereal, coffee grounds, cleaning powder, etc.) may be poured from the flexible enclosure **100**. To close the access opening **150**, opposite portions of the spout **152** are moved toward each other. For example, the user may pinch or squeeze opposing sections of the first side wall **106** that form the spout **152**. The resealer **134** located along an interior surface of the first side wall **106** may securely close the access opening **150**. For instance, two opposing fastener strips may engage one another. The resealer **134** may be self-mating such that the fastener strips have a common structured surface that is capable of mating with itself. As discussed above, however, the fastener strips may have different structures and be of different types.

(31) FIG. **5** is a schematic diagram of a system **200** formed in accordance with an embodiment, and FIG. **6** is a flowchart of a method **300** that may be carried out by the system **200**. The system **200** may be used in the manufacturing of a flexible enclosure, such as the flexible enclosure **100** (FIG. **1**). It should be noted that the particular arrangement of components (e.g., the number, types, placement, or the like) of the illustrated embodiment may be modified in various alternate embodiments. In various embodiments, different numbers of a given element may be employed, a different type or types of a given element may be employed, a given element may be added, or a given element may be omitted.

(32) The system **200** includes a plurality of actuators that move continuous webs of flexible material. For example, the actuators may include a first actuator **202**, a second actuator **204**, and a third actuator **206**. The actuators may be rotary actuators or servo-motors. As shown in FIG. **5**, the

first actuator **202** is operably connected to a roll or spool **213** of a first continuous web **212** of flexible material, the second actuator **204** is operably connected to a roll or spool **215** of a second continuous web **214** of flexible material, and the third actuator **206** is operably connected to a roll or spool **217** of a third continuous web **216** of flexible material. The flexible material of each web may comprise a single material or may be a composite material comprising two or more layers of different materials.

(33) The method **300** includes providing, at **302** (FIG. 6), the first continuous web **212** moving in a first flow direction **222**. For example, the first actuator **202** may cause the roll **213** of the first continuous web **212** to unwind, thereby moving the first continuous web **212** in the first flow direction **222**.

(34) In some embodiments, the first continuous web **212** is provided to a trans-directional (TD) applicator **208**. As the continuous web **212** moves in the first flow direction **222**, the TD applicator **208** may apply, at **304** (FIG. 6), a resealer **230**. In particular embodiments, the resealer **230** includes a fastener strip (e.g., zipper strip). The resealer **230** may extend in a transverse direction **225** that is perpendicular to the first flow direction **222**. In some embodiments, a line of weakness and/or slit (generally referred to by **234**) may be applied by the TD applicator **208** or by an adjacent system. The line of weakness and/or slit **234** may extend adjacent to and parallel to the resealer **230**.

(35) In the illustrated embodiment, the resealer **230** and the line of weakness and/or slit **234** are applied to the first continuous web **212** as the first continuous web **212** to the next manufacturing stage. In other embodiments, at least one of the resealer **230** or the line of weakness and/or slit **234** are pre-applied. For example, the first continuous web **212** may be unwound and the resealer **230** and the line of weakness and/or slit **234** applied. The first continuous web **212** may then be rewound and transported to another location or stored for later use.

(36) At **306** (FIG. 6), a first segment **242** of the first continuous web **212** is folded into a desired shape. At **308** (FIG. 6), the first segment **242** is separated from the first continuous web **212**. To fold the first segment **242**, one or more surfaces of an apparatus **218** may shape the first segment **242** as the first segment **242** glides along the surface(s) of the apparatus **218**. For example, the apparatus **218** may include a funnel-shaped tube that is axially aligned with the first continuous web **212** and the first flow direction **222**. As the first continuous web **212** flows through the tube, the surfaces of the tube force the first continuous web **212** into a desired shape. As shown, the folded first segment **242** may have an oval shape or may be C-shaped. For example, opposite edges of the first segment **242** may be adjacent to each other with a small gap therebetween, may essentially contact each other, or may overlap each other.

(37) The first segment **242** may then be removed from the remaining body of the first continuous web **212** using, for example, a cutting machine **210**. The cutting machine **210** may mechanically separate (or cut) the flexible material using, for example, a blade. However, other mechanisms may be used to separate the first segment **242** from the remaining body of the first continuous web **212**.

(38) In the illustrated embodiment, the first segment **242** is folded and then separated from the first continuous web **212**. In other embodiments, the first segment **242** may be separated from the first continuous web **212** and then folded into the desired shape. In some embodiments, the cutting machine **210** and the apparatus **218** may be essentially part of the same machine or sub-system.

(39) At **310** (FIG. 6), the folded first segment is attached to a segment of the second continuous web **214** and a segment of the third continuous web **216**. Segments may be attached by sealing portions of two or more segments to each other. For example, the folded first segment **242** may be attached to the second continuous web **214** and the third continuous web **216** through thermal sealing in which thermal energy (heat) and pressure cause a seal to form along the joined segments. A common method of sealing two or more segments to each other is heat-sealing in which heated bars press the segments to each other causing the layers to melt and the two segments to bond or fuse together. Another method of sealing two or more segments to each other may include ultrasonic sealing. Other methods of sealing may include impulse sealing, band sealing, and hot

wire sealing.

(40) In some embodiments, top sealed regions, and bottom sealed regions for one side of the flexible enclosure may be formed when the first folded segment **242** and the third continuous web **216** undergo a heat-sealing process. Top sealed regions and bottom sealed regions for the other side of the flexible enclosure may be formed when the first folded segment **242** and the second continuous web **214** undergo a heat-sealing process. These two processes may be performed simultaneously or at different times. In FIG. 5, it appears that the third continuous web **216** and the first folded segment **242** are attached to each other prior to the second continuous web **214** being attached to the first folded segment **242**. In other embodiments, the second continuous web **214** may be attached before the third continuous web **216** is attached. Alternatively, the second continuous web **214** and the third continuous web **216** may be concurrently attached to the first folded segment **242**.

(41) The first and second continuous webs **214**, **216** with one or more first folded segments **242** positioned therebetween may be referred to as a composite web from which precursor enclosures may be obtained. At **312** (FIG. 6), a cutting machine **248** may separate the precursor enclosures from the composite web. FIG. 5 also shows an example cross-section of a precursor enclosure **250**. The precursor enclosure **250** includes one section **252** of a first folded segment (segment A) and another section **254** of a different first folded segment (segment B). The section **254** includes a resealer, but the section **252** does not include a resealer. As such, in the illustrated embodiment, the first folded segment **242** is cut in half such that one portion functions as a first side wall for one precursor enclosure and the other portion functions as a second side wall for a subsequent precursor enclosure. At **314** (FIG. 6), other openings may be sealed and the flexible enclosure may be filled with a desired product.

(42) FIG. 7 is a cross-section of a three-sided flexible enclosure **400**, and FIG. 8 is a schematic illustration of a system **450** that may be used to make the flexible enclosure **400**. The flexible enclosure **400** may include elements and features that are similar or identical to the elements and features of the flexible enclosure **100**. The system **450** may include elements and features that are similar or identical to the elements and features of the system **200**.

(43) The flexible enclosure **400** includes a front wall (or first wall) **402**, a back wall (or second wall) **404**, and only a single side wall **406**. The walls **402**, **404**, **406** are discrete sections of a flexible material. For example, the front wall **402** may be obtained from a first continuous web **412** (FIG. 8) of flexible material, the back wall **404** may be obtained from a second continuous web **414** (FIG. 8) of flexible material, and the side wall **406** may be obtained from a third continuous web **416** (FIG. 8) of flexible material. As shown in FIG. 7, the front wall **402** and the back wall **404** may be directly attached to each other through a heat-seal **405**. The heat-seal **405** forms a side edge of the flexible enclosure **400**. The front wall **402** and the back wall **404** may be directly attached to opposite edges of the side wall **406** through heat seals **407**, **409**.

(44) Also shown, the side wall **406** includes a resealer **434** that is positioned along an interior surface of the side wall **406**. Optionally, a line of weakness (not shown) may be provided to permit a user to create a slit along the resealer **434**. Similar to the flexible enclosure **100** (FIG. 1), the slit permits a user to pull a bottom portion of the side wall **406** outwardly and form a spout. The resealer **434** may be used to close the spout.

(45) As shown in FIG. 8, the resealer **434** extends transverse to a flow direction of the first continuous web **412**. The resealer **434** extends across an entire width of the first continuous web **412**. A folded segment **442** of the first continuous web **412** may be folded and separated from the first continuous web **412**. The folded segment **442** is C-shaped. Unlike the folded segment **242** (FIG. 5), the folded segment **442** will form only one side of a flexible enclosure. Opposite edges of the folded segment **442** are attached to the second and third continuous webs **414**, **416**. The second and third continuous webs **414**, **416** may then be directly attached to each other along the heat seal **405**, which forms the side edge of the flexible enclosure **400**. After attaching the second and third

continuous webs **414**, **416**, the composite web may be sliced along the heat seal **405** to separate the flexible enclosure **400** from the composite web.

(46) As shown in FIGS. **7** and **8**, the C-shaped side wall **406** is concave such that the side wall **406** is shaped inwardly and has the resealer **434** along an interior surface of the side wall **406**. In alternative embodiments, the side wall is C-shaped but convex such that the side wall extends outwardly. The resealer may be positioned along an interior surface of the side wall as described above. In such embodiments, the side wall may be shaped like a spout prior to opening.

(47) FIG. **9A** is a cross-section of a closed three-sided flexible enclosure **500**, and FIG. **9B** is a cross-section of the three-sided flexible enclosure **500** when opened. FIG. **10** is a schematic illustration of a system **550** that may be used to make the flexible enclosure **500**. The flexible enclosure **500** may include elements and features that are similar or identical to the elements and features of the other flexible enclosures described herein. The system **550** may include elements and features that are similar or identical to the elements and features of the other systems described herein.

(48) The flexible enclosure **500** includes a first wall (or front wall) **502**, a second wall (or back wall) **504**, and only a single side wall **506**. The walls **502**, **504**, **506** are discrete sections of a flexible material. For example, the first wall **502** may be obtained from a first continuous web **512** (FIG. **10**) of flexible material, the second wall **504** may be obtained from a second continuous web **514** (FIG. **10**) of flexible material, and the side wall **506** may be obtained from a third continuous web **516** (FIG. **10**) of flexible material. As shown in FIG. **9B**, the first wall **502** and the second wall **504** may be directly attached to each other through a heat-seal **505**. The heat-seal **505** forms a side edge of the flexible enclosure **500**. The first wall **502** and the second wall **504** are directly attached to opposite edges of the side wall **506** through heat seals **507**, **509**.

(49) Unlike the flexible enclosures **100**, **400**, which have side walls with resealers, the side wall **506** does not have a resealer. Instead, the first wall **502** has a resealer **534** positioned along an interior surface of the first wall **502**. To provide the resealer **534** along the first wall **502**, the resealer **534** is applied to the first continuous web **512**. More specifically, the resealer **534** is applied in a direction that is parallel to the flow direction of the first continuous web **512**. The flexible enclosure **500** may then be manufactured in a manner that is similar to the flexible enclosure **400**.

(50) In the illustrated embodiment of FIGS. **9A**, **9B**, and **10**, the first wall **502** is a front wall. In other embodiments, the first wall **520** may be a back wall or, alternatively, the resealer may be applied to the second wall, which may function as a front wall or a back wall.

(51) In some embodiments, the resealer **534** is a two-piece resealer that includes a first segment **534A** (FIG. **9B**) and a second segment **534B** (FIG. **9B**) that are configured to mate with each other. For example, the first and second segments **534A**, **534B** may be zipper segments having male and female interlocking features. The first segment **534A** may be coupled to a bottom wall portion of the first wall **502**, and the second segment **534B** may be coupled to a top wall portion of the first wall **502**.

(52) FIG. **11** is a cross-section of a four-sided flexible enclosure **600**, and FIG. **12** is a schematic illustration of a system **650** that may be used to make the flexible enclosure **600**. The flexible enclosure **600** may include elements and features that are similar or identical to the elements and features of the other flexible enclosures described herein. The system **650** may include elements and features that are similar or identical to the elements and features of the other systems described herein.

(53) The flexible enclosure **600** includes a front wall (or first wall) **602**, a back wall (or second wall) **604**, and first and second side walls **606**, **608**. The walls **602**, **604**, **606**, **608** are discrete sections of a flexible material. For example, the first and second side walls **606**, **608** may be obtained from a first continuous web **612** (FIG. **12**) of flexible material, the front wall **602** may be obtained from a second continuous web **614** (FIG. **12**) of flexible material, and the back wall **604**

may be obtained from a third continuous web **616** (FIG. 12) of flexible material.

(54) The flexible enclosure **600** includes a resealer on an interior surface of each of the walls **602**, **604**, **606**, **608**. To provide the sealers, each of the first, second, and third continuous webs **612**, **614**, **616** has a resealer applied thereto. For example, the first continuous web **612** may have a series of spaced-apart of sealers **661** applied along a width of the first folded segment. The sealers **661** extend in a direction that is transverse to the flow direction of the first continuous web **612**. The second and third continuous webs **614**, **616** may have sealers **662**, **663**, respectively, applied thereto in a direction that is parallel to the flow direction. The flexible enclosure **600** may be manufactured in a manner that is similar to the flexible enclosure **100**.

(55) As shown in FIG. 11, the flexible enclosure **600** has a plurality of sealers **661-663** that, as a group, extend entirely around or nearly entirely around an interior surface of the flexible enclosure **600**. Each sealer may align with the other sealers such that an entire top portion of the enclosure may be removed. After opening, the flexible enclosure **600** may be closed by pressing the opposite front and back walls **602**, **604** toward each other. For example, the sealers **662**, **663** of the front and back walls **602**, **604**, respectively, may engage each other. The sealer **661** along the side wall **608** may engage itself and/or the sealer **661** along the side wall **606** may engage itself. Optionally, portions of the sealers **662**, **663** of the front and back walls **602**, **604** may engage the sealers **661** along the side walls.

(56) It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the inventive subject matter without departing from its scope. While the dimensions and types of materials described herein are intended to define the parameters of the inventive subject matter, they are by no means limiting and are example embodiments. Many other embodiments will be apparent to one of ordinary skill in the art upon reviewing the above description. The scope of the inventive subject matter should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

(57) This written description uses examples to disclose several embodiments of the inventive subject matter and also to enable one of ordinary skill in the art to practice the embodiments of inventive subject matter, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the inventive subject matter is defined by the claims, and may include other examples that occur to one of ordinary skill in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

(58) As used herein, an element or step recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural of said elements or steps, unless such exclusion is explicitly stated. Furthermore, references to “one embodiment” of the present inventive subject matter are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, embodiments “comprising,” “including,” or “having” an element or a plurality of elements having a particular property may include additional such elements not having that property.

Claims

1. A flexible enclosure comprising: a front wall and a back wall that oppose each other and a side wall that joins the front and back walls, wherein the front wall, the back wall, and the side wall are discrete sections, the front wall and the back wall being attached to opposite side edges of the side wall, the side wall having a slit between top edges and bottom edges of the front wall and the back wall to define a top wall portion of the side wall between the slit and the top edges of the front wall and the back wall and to define a bottom wall portion of the side wall between the slit and the bottom edges of the front wall and the back wall; and a resealer extending between the front and back walls along the bottom wall portion of the side wall, wherein the bottom wall portion is separable from the top wall portion along the slit, the bottom wall portion configured to flex outwardly and away from the top wall portion and form a spout of the flexible enclosure, the resealer configured to close the spout; wherein the bottom wall portion is configured to flex outwardly and away from the top wall portion to form the spout and the resealer is configured to close the spout while the bottom wall portion is outwardly flexed away from the top wall portion, wherein the side wall is a first side wall and the flexible enclosure includes a second side wall that is also a discrete section with respect to the front wall, the back wall, and the first side wall, the front wall and the back wall being attached to opposite edges of the second side wall, the first and second side walls opposing each other, wherein the resealer is one of four rescalers, each of the rescalers extending along a respective one of the first side wall, the second side wall, the front wall, and the back wall.
2. The flexible enclosure of claim 1, wherein the side wall is configured to have a line of weakness, the slit being formed when the line of weakness is broken.
3. The flexible enclosure of claim 1, wherein the side wall is a gusseted side wall.
4. The flexible enclosure of claim 1, wherein the resealer includes a self-mating strip extending between the front and back walls along an interior surface of the side wall.
5. The flexible enclosure of claim 4, wherein the self-mating strip has opposite crushed ends and each edge of the side wall has one of the crushed ends.
6. The flexible enclosure of claim 1, wherein the resealer includes first and second segments that are configured to mate with one another to close the spout.
7. The flexible enclosure of claim 1, wherein the resealer is a first resealer and the second side wall has a second resealer that extends between the front and back walls, the second side wall configured to have a slit between top and bottom wall portions of the second side wall, wherein the second resealer extends between the front and back walls along the bottom wall portion of the second side wall, wherein the bottom wall portion of the second side wall is separable from the top wall portion along the slit, the bottom wall portion of the second side wall being configured to flex outwardly and away from the top wall portion, the resealer of the second side wall configured to close the second side wall.
8. The flexible enclosure of claim 7, wherein the front and back walls have respective slits and respective rescalers and wherein the slits of the first and second side walls and the front and back walls enable a user to remove a top of the flexible enclosure to provide an open-ended enclosure having an open top, the rescalers of the first and second side walls and the front and back walls enabling a user to close the open top of the open-ended enclosure.
9. The flexible enclosure of claim 1, wherein the top wall portion remains in a stationary position when the spout is in an open position and when the spout is in a closed position.
10. The flexible enclosure of claim 1, wherein the resealer is configured to close the spout while the top wall portion is positioned inwardly of the resealer.
11. A flexible enclosure comprising: a front wall and a back wall that oppose each other and a first side wall that joins the front and back walls, wherein the front wall, the back wall, and the first side wall are discrete sections, the front wall and back wall being attached to opposite side edges of the first side wall, the first side wall having a first slit between top edges and bottom edges of the front

wall and the back wall to define a top wall portion of the side wall between the first slit and the top edges of the front wall and the back wall and to define a bottom wall portion of the first side wall between the slit and the bottom edges of the front wall and the back wall; a first resealer extending between the front and back walls along the bottom wall portion of the first side wall, wherein the bottom wall portion is separable from the top wall portion along the first slit, the bottom wall portion configured to flex outwardly and away from the top wall portion and form a spout of the flexible enclosure, the first resealer configured to close the spout; wherein the bottom wall portion is configured to flex outwardly and away from the top wall portion to form the spout and the first resealer is configured to close the spout while the bottom wall portion is outwardly flexed away from the top wall portion; and a second side wall that is also a discrete section with respect to the front wall, the back wall, and the first side wall, the front wall and the back wall being attached to opposite edges of the second side wall, the first and second side walls opposing each other, the second side wall having a second resealer that extends between the front and back walls, the second side wall configured to have a second slit between top and bottom wall portions of the second side wall, wherein the second resealer extends between the front and back walls along the bottom wall portion of the second side wall, wherein the bottom wall portion of the second side wall is separable from the top wall portion along the second slit, the bottom wall portion of the second side wall being configured to flex outwardly and away from the top wall portion, the second resealer of the second side wall configured to close the second side wall.

12. The flexible enclosure of claim 11, wherein the first side wall is configured to have a line of weakness, the first slit being formed when the line of weakness is broken.

13. The flexible enclosure of claim 11, wherein the first side wall is a gusseted side wall.

14. The flexible enclosure of claim 11, wherein the first resealer includes a self-mating strip extending between the front and back walls along an interior surface of the first side wall.

15. The flexible enclosure of claim 14, wherein the self-mating strip has opposite crushed ends and each edge of the first side wall has one of the crushed ends.

16. The flexible enclosure of claim 11, wherein the first resealer includes first and second segments that are configured to mate with one another to close the spout.

17. The flexible enclosure of claim 11, wherein the front and back walls have respective third slits and respective third sealers and wherein the first and second slits of the first and second side walls and the third slits of the front and back walls enable a user to remove a top of the flexible enclosure to provide an open-ended enclosure having an open top, the first and second sealers of the first and second side walls and the front and back walls enabling a user to close the open top of the open-ended enclosure.

18. The flexible enclosure of claim 11, wherein the first resealer and the second resealer are two of four sealers, each of the sealers extending along a respective one of the first side wall, the second side wall, the front wall, and the back wall.

19. The flexible enclosure of claim 11, wherein the top wall portion remains in a stationary position when the spout is in an open position and when the spout is in a closed position.

20. The flexible enclosure of claim 11, wherein the first resealer is configured to close the spout while the top wall portion is positioned inwardly of the first resealer.
